Rethinking Evidence Selection in Multimodal Fact-Checking: Calibrated Adaptive Set Optimization Beyond Fixed Top-k Retrieval

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***Abstract*—To perform evidence-based multimodal fact-checking, an evidence retrieval system is needed before making a verdict, to retrieve identifiable text and image evidence. A typical design uses the same fixed top-k evidence budget for every claim, although the amount of available evidence may vary, and the precision–recall trade-off due to retrieval imperfection is also not taken into account. We study if instead evidence-set size can be selected by directly optimizing the expected set quality. We present CEFES (Calibrated Expected-F1 Evidence Selection) to estimate calibrated evidence-membership probabilities and select the topmost ranked prefix with maximum exact expected set-F1 under an explicit independent-Bernoulli approximation. By preserving candidates, frozen features, scorer inputs, and the downstream verifier, the selection rule is the controlled difference between adaptive and fixed-budget selectors. The evidence F1 for CEFES on a checked judgment-candidate packet with 400 claims is 0.0683 higher than that of the training-calibration-selected fixed budget, although a calibrated threshold is stronger. This feasibility result flips under label-blind retrieval from all 91,822 released text documents and 90,270 train/development image paths: CEFES is on average 0.0418 evidence F1 worse than the selected fixed budget in every seed, and each paired 95% interval is negative. A matched count-classification control is, on average, 0.0185 evidence F1 higher than CEFES. The failed primary evidence criterion takes precedence over the shared-verifier diagnostic, which has a macro-F1 difference of 0.0577 in favor of CEFES, on average. Independent audits and a clean rerun of the retrieval and method records are congruent. Thus, the prespecified development gate is not met, the official MOCHEG test remains unopened, and the study identifies candidate construction and retrieval quality as important conditions for evaluating adaptive evidence selection instead of claiming superiority.**

***Index Terms*—multimodal fact checking, evidence selection, calibrated retrieval, expected F1, selective prediction**

I. Introduction

There are several approaches to automated fact-checking, many of which can be implemented as a pipeline: finding a claim that can be fact-checked, finding potential evidence sources, evaluating evidence and predicting a verdict [1], [2]. Evidence-based systems are distinct from classifiers that rely on content alone as they require that the output be backed up by identifiable source material rather than only the linguistic style, the publisher or the appearance of the post. This distinction implies that retrieval and evidence attribution are part of the scientific claim and that a correct verdict is not fully justified when the cited material is irrelevant or incomplete.

When claims and evidence are multimodal, it is more difficult. A claim can combine a caption and an image, while decisive evidence may be distributed across a fact-check article, a quoted passage and one or more images. The modalities can be complementary, redundant or contradictory and a system needs to uphold the identity of every item selected if the reason for the selection is to be audited. MOCHEG materialises this situation as joint text–image evidence retrieval, three-way claim verification, and explanation generation [3]. Subsequent systems incorporate dense retrieval, visual language, language-model verification, or graph-based evidence reasoning [4], [5]. These developments enable us to make evidence selection a measurable component, a step away from the invisible preprocessing step.

The output of a correct verdict and useful evidence set are different but related. A model may produce a correct label, and yet mention irrelevant items or fail to mention an important image or produce many redundant documents. Fluently-generated explanations can also be inaccurate in attributing their statements to the evidence chosen [6], [7]. Therefore, the evaluation should maintain the identity of the items selected and be based on how much they overlap with the released gold evidence and not be based on a verdict or a plausible explanation.

Most multimodal retrieval systems still present a fixed (universal) set of top-k results. A single budget is easy to reproduce and use, but it assumes that each claim requires the same number of items. The higher the value of k, the more gold evidence that can be recovered but with less precision and vice versa. When the ranking is not perfect, predicting just the gold count is not enough to answer this trade-off: knowing how many items are needed does not identify which ranked items are correct. A previous SciVer study in this project found a negative result: a learned count head improved count accuracy but did not beat the strongest fixed budget in evidence-set F1. The above-mentioned completed benchmark is only used as motivation and provides no selection or confirmation evidence for the current study.

The question in this study is whether it is possible to target evidence-set utility directly by the selection rule. Calibrated Expected-F1 Evidence Selection (CEFES) estimates the probability of each candidate being in the gold evidence set, sorts the

candidates by these probabilities and checks each prefix that is admissible. An exact Poisson-binomial dynamic program is used to calculate expected set-F1 under an explicitly stated independent-Bernoulli approximation and the decoder outputs the smallest prefix where the expectation is maximized. For all outputs (fixed-budget, threshold, count-prediction and CEFES), the same candidate packet, frozen features and scorer family are used; and for end-to-end comparison, the same verifier checkpoint is used. The proposed solution consequently alters the set decision, but preserves retrieval access and the capacity of downstream models.

The evaluation is deliberately split into a convenient candidate universe and realistic retrieval. The checked 400-claim feasibility packet is a first test of the selector on released judgment candidates ranked without relevance labels. The primary development comparison then retrieves label-blind from all 91,822 released text documents and 90,270 train/development image paths. The two stages include pinned BGE-M3 and SigLIP2 features, as well as three seeds, matched controls, paired bootstrap analysis and independently recomputed artifacts. CEFES outperforms the chosen fixed budget in the restricted packet, but is outperformed under full-corpus retrieval in all seeds. As a result, the main development gate fails and the official test, which has not been touched, is still sealed.

This development study's contribution is:

- a multimodal evidence selector which adaptively optimizes the exact expected set-F1 instead of the number of citations or a global retrieval depth;
- a matched comparison protocol, where the same set of candidates, set of representations, inputs by the scorer, accesses to tuning, and checkpoints by the verifier are shared between the adaptive and fixed selectors;
- an evaluation, calibration and evidence-count test-sealed MOCHEG development;
- an auditable development gate to keep the official test from being used when the proposed selector fails to create a clear development benefit; and
- empirical data showing that a selector conclusion extracted from a judgment-candidate packet can reverse under label-blind full-corpus retrieval.

So the paper does not claim empirical superiority. It finds that the initial positive result for adaptive evidence selection on the packet does not hold in the end-to-end development environment because of candidate construction and retrieval error. Reporting this reversal, including the failed seeds and preserved artifacts, is important for distinguishing a selector mechanism result from a realistic retrieval result.

II. Related Work

A. Automated Fact-Checking and Evaluation Resources

Recent surveys differentiate fake-news classification from evidence-based fact-checking: The former is to predict the veracity of a post and its context; the latter is normally decomposed into a series of tasks, namely claim analysis, evidence retrieval, evidence assessment, and verdict prediction [1], [2], [8]. Review of datasets indicate that the reported performance is dependent on the domain and the construction of labels, as well as the modalities covered and controls for event or source overlap [9], [10], [11]. Despite this, there are still persistent challenges such as cross-domain generalization, incomplete explanations, and inconsistent benchmark protocols [12], [13], [14]. These differences are important as CEFES is assessed as an evidence-selection and verification system, not just as a content classifier.

While sentence-level scientific evidence is required for SciFact, multi-document evidence is needed for HoVer, evidence needed to address claims in real-world contexts for COVID-Fact, and multilingual variation in X-Fact [15], [16], [17], [18]. External-document systems can mitigate the necessity to rely on stylistic shortcuts, but they can also raise retrieval, freshness, and evidence-selection mistakes [19], which led us to measure verdict prediction and evidence attribution separately.

B. Multimodal Misinformation Detection

Content-based multimodal detectors learn the interaction between text and image with bilinear pooling [20], with hierarchical attention [21], by incorporating topic memory [22] or using cross-attention [23] or ambiguity modeling [24] or with multi-level fusion [25]. Following systems incorporate contrastive, within-modality, stylometric and semantic-alignment signals [26], [27], [28], [29]. Their main target, however, is still a veracity label at the post level and not a set of evidence that is identifiable.

The mismatch between local and global information, attention mechanism with matching, cluster heterogeneity, semantic-space alignment, and conflicting unimodal signals have been studied recently [30], [31], [32], [33], [34]. These methods can enhance the cross-modal representations, but a fused latent vector does not specify which item of concrete evidence can be used to support the verdict.

There are out-of-context benchmarks like NewsCLIPpings [35], COSMOS [36] and VERITE [37] which test misleading image–caption combinations and control unimodal bias. DGM4, SNIFFER, MMDFND, and IMFND further enhance localization, explanation, domain coverage or use of large vision–language models [38], [39], [40], [41]. In the present study, the auditable attribution unit is taken as the document evidence, which is provided in the dataset.

C. Multimodal Evidence Retrieval and Verification

The MOCHEG project has created a shared benchmark for multimodal evidence retrieval, verdict prediction and explanation generation [3]. M-RAV uses dense retrieval, image–text consistency verbalization, relevance estimation and language-model verification [4] while MEVER constructs a text–image evidence graph and conducts token- and evidence-level fusion [5]. Related retrieval studies distinguish evidence utility from semantic similarity and learn when visual retrieval is necessary [42], [43]. Overall, it is clear that retrieval and fusion are significant aspects of multimodal fact-checking. However, their evidence output is typically controlled by a global retrieval depth or generation procedure rather than an explicitly evaluated per-claim citation budget.

TABLE I
COMPARISON OF MULTIMODAL EVIDENCE-AWARE STUDIES. "ADAPTIVE" DENOTES VARIABLE-SIZE EVIDENCE-SET OUTPUT; RERANKING, FILTERING, AND MODALITY ROUTING ARE NOT ADAPTIVE SET DECODING.

Study	Setting	Retrieval/selection mechanism	Set policy	Evaluated outputs
MOCHEG [3]	Web fact-checking	Separate neural text/image retrieval	Fixed top-k	Retrieval, verdict, explanation
RED-DOT [44]	VERITE; NewsCLIPpings+	Relevance detection and evidence reranking	Fixed pool with item masking	Item relevance, verdict
SciVer [45]	Scientific papers	Foundation-model and RAG benchmark	No adaptive set policy	Evidence, verdict
MuSciClaims [46]	Scientific figures	Figure localization and reasoning diagnostics	Provided figure context	Localization, verdict
M^2-TabFact [47]	Multi-document tables	Textual/visual table evidence reasoning	Provided multimodal evidence	Verdict
M-RAV [4]	MOCHEG; FACTIFY; Fin-Fact	BGE/CLIP retrieval, alignment, relevance scoring	Tuned global top-k	Verdict with gold/retrieved evidence
MEVER [5]	MOCHEG; FACTIFY; AIChartClaim	Two-layer graph retrieval and multimodal fusion	Fixed top-k	Retrieval, verdict, explanation
Utility-oriented [42]	MRAG-Bench; Visual-RAG	Information-gain/helpfulness ranking	Fixed top-K	Downstream answer utility
Selective retrieval [43]	AVerImaTeC	Question-type routing with hybrid/SigLIP retrieval	Fixed per-question budget	Evidence, verdict, justification
CEFES (ours)	**MOCHEG development**	**Calibrated membership scoring and exact expected-F1 decoding**	**Adaptive prefix**	**Set F1; verifier diagnostic**

Moreover, RED-DOT is particularly relevant because it detects and reranks relevant external evidence before verdict prediction [44]. Its item-relevance mechanism differs from choosing a variable-size evidence set against a released set-level target. These operations—fixed retrieval, relevance filtering, modality routing, utility ranking and adaptive set decoding—are not considered interchangeable in Table I.

D. Scientific Multimodal Claim Verification

In scientific writing, claims can be associated with stable locations such as passages, tables and figures. SciVer is a collection of 3,000 claim–evidence instances annotated by experts from 1,113 papers, which distinguishes multiple types of reasoning [45]. MuSciClaims focuses on figure-grounded claims and controlled perturbations of the claims that reveal the shortcomings of evidence localization and aggregation across modalities [46]. M^2-TabFact also supports verifying tables presented in images or text throughout multiple documents [47]. While these resources are helpful for item-level evidence evaluation, which is challenging to do in open-web misinformation, they are also constrained to the scientific domain and thus cannot be directly applied to political or breaking-news claims.

E. Evidence Attribution and Faithful Rationales

An explanation may be understandable while citing the wrong source or reflecting a decision based on another signal. Early research on fact-checking explanation generation focused on generating text-based explanations that would be helpful [48]. Previous work used evidence rationales as an explicit evaluation target [49] and faithful-by-construction rationalizers only allowed predictors to attend to a subset of evidence spans [50]. Citation-grounded language generation assesses if the generated statements are backed up by retrieved information [51] and RAGTruth is a record of hallucinated statements that are still possible in retrieval-augmented output [7]. At the most basic level, citation masking and recovery evaluates whether a fact-checking explanation cites the right evidence [6].

F. Position of This Study

The literature studied shows strong models to make a verdict, fusion mechanisms for multimodal systems, pipelines for retrieval, explanation generators and citation metrics, but evidence output is often still determined by a global retrieval depth. Utility-oriented and selective multimodal retrieval step towards the current question by judging the usefulness of visual evidence [42], [43], but they do not consider exact expected set-F1 decoding for a variable text–image evidence set under a matched fact-checking verifier. The most similar task and system context are offered by MOCHEG, M-RAV and MEVER [3], [4], [5].

This is the narrower gap that CEFES targets. It calibrates evidence-membership probabilities at the item level and chooses the prefix which maximizes exact expected set-F1, while fixed-budget, threshold and count controls share candidates and representations. The novelty claim is limited to this selection objective, decoder and controlled evaluation protocol. The study is not intended to offer novel contributions to multimodal encoders, general graph reasoning, evidence retrieval or explanation generation, and it does not claim empirical superiority.

III. Task and Data

A. Task Definition

For claim c and public multimodal corpus D, retrieval returns a bounded set of text and image candidates $E = \{e_1, \ldots, e_n\}$. Every candidate possesses a permanent, modality-qualified identifier, modality, content or media path, retrieval features, and a binary released relevance judgment if the split allows for supervision. The system returns an evidence set $S \subseteq E$ and a verdict of SUPPORTED, REFUTED or NOT ENOUGH INFORMATION. Only evidence released with an identifier matching an official gold item under the benchmark's evaluation rule is correct. The generation of explanations is beyond the scope of the main contribution.

Given a predicted set S and a gold set G, claim-level evidence F1 is

$$F_1(S, G) = \begin{cases} 1, & |S| + |G| = 0, \\ \dfrac{2|S \cap G|}{|S| + |G|}, & \text{otherwise.} \end{cases} \tag{1}$$

The primary selector endpoint is the mean of this claim-level score. Verdict macro-F1 is a co-primary operational safeguard because a selector is not helpful if it results in a significant decrease in verification performance.

B. Evidence Reachability and Evaluation Regimes

Candidate construction limits selection quality. Let E_c be the packet that is exposed to a selector for claim c and G_c be the complete released gold set. The gold set that can be reached is $G_c^E = G_c \cap E_c$. An oracle that selects all reachable gold items and no distractors is a packet-conditioned ceiling for a nonempty gold set

$$F_1^{\text{ceiling}}(c) = \frac{2|G_c^E|}{|G_c| + |G_c^E|}. \tag{2}$$

This maximum is 1 only if the packet has the complete gold set. It decreases whenever retrieval misses a gold item, irrespective of the configuration of the downstream scorer and the behavior of the set decoder. The ceiling is not a deployable result, but is diagnostic in that it relies on the released relevance judgments following retrieval and assumes that all reachable gold items are identified.

This separation leads to two evaluation regimes. A good judgment-candidate packet is one that contains a "dense" set of assessed positives and negatives with stable identifiers, good for a controlled mechanism test. It is not a measure of whether or not the system can also find those candidates in the public corpus. A label-blind full-corpus packet, on the other hand, involves the full chain, from corpus retrieval to evidence selection. Relevance labels are only added after the ranking is frozen, and evidence missing from the packet remains a false negative in Eq. 1. Hence, the result from the judgment packet is reported as feasibility evidence and the full-corpus comparison is considered the main development decision.

TABLE II
OFFICIAL MOCHEG SPLIT STATISTICS REPORTED BY THE DATASET PAPER.

Split	Claims	Text evidence	Image evidence
Train	11,669	23,545	8,927
Development	1,490	4,067	1,178
Test	2,442	6,268	2,007

C. Public MOCHEG Release

MOCHEG provides end-to-end multimodal fact-checking data under CC BY 4.0 and code under Apache 2.0 [3]. Table II summarizes the official statistics: 11,669 train claims, 1,490 development claims and 2,442 test claims, along with 91,822 corpus documents and 122,246 images. The labels are Supported, Refuted and Not Enough Information.

The authoritative archive is kept as the recovery source, and the development process is kept in step with the authoritative release by using the public transport schemas and counts as the basis for development. Complete immutable revisions for the train, development, shared-corpus and sealed-test transports are documented in the released configuration and study plan. Test annotations are not downloaded during the development process.

D. Development Packet

The current feasibility packet contains 200 train and 200 development claims with a deterministic (hard-coded) seed. The sentence preprocessing is based on the public 91,822-row corpus which is used to reconstruct the candidate text. Only candidate images are downloaded, and those images are audited for byte identity and Pillow readability. Modality-qualified judgment rows are only collapsed if they have the same relevance label; otherwise, the conflict is considered a release error.

The packet includes 15,823 candidate records, of which 15,289 are available. There is no gold evidence among the missing records: all missing records are negative distractors, so all retained claims have their released gold evidence. The 1,612 downloaded image files contain 1,255 unique byte hashes, with 357 exact duplicate occurrences. These counts are indicators of packet integrity, but not model performance. In the feasibility experiment, the released judgment-candidate universe is ranked using label-blind semantic similarity and at most 40 text and 20 image items are retained. A positive result among candidates drawn from this restricted universe must be replicated with candidates drawn from the full released corpus.

E. Split Integrity and Test Seal

Train data are split into fitting and calibration folds and only train and official development are used for all configuration selection. Development code refuses test annotations. A formal run requires a frozen protocol containing code, configuration, dataset, feature and checkpoint identities, together with an explicit formal-test flag. Data preparation, feature extraction and pilot execution may not print any test metric. Once a method

has been formally evaluated and then changed, the existing test result can no longer be presented as fresh confirmation.

Data audits cover changes to the data sources, release counts, labels, hashes, missing media, exact duplicates, perceptual image duplicates, and any grouping of sources or events available from the release. The allowed training fold is the only one on which learned preprocessing and calibration are done. The complete access and identity audit is kept outside the manuscript in the research records of the project.

IV. Methodology

A. Overview

CEFES separates candidate retrieval, evidence-membership estimation, calibration, set decoding and verdict prediction. All the controlled selectors of Fig. 1 get the same bounded packet and feature artifact. The adaptive rule only changes which evidence identifiers to emit and receives no additional corpus, label or tuning access. The shared verifier is a secondary diagnostic and is not used to override a failed evidence-selection gate.

B. Frozen Multimodal Features

The text branch is based on BGE-M3 at the immutable revision used for the development configuration. Before calculating cosine similarity, mean-pooled, attention-masked claim and evidence embeddings are normalized. The image branch is based on SigLIP2-base-patch16-224 in the same manner. Text and recoverable images are encoded individually and normalized. Both models are frozen, cached and shared amongst seeds and selector families.

The pilot feature vector implemented for claim–candidate pair (c, e_i) is

$$\begin{aligned} x_i = [&s_i, s_i^2, m_i^t, m_i^v, \\ &\log(1+\ell_i)/10, \log(1+\ell_c)/10, \\ &\log(1+n_c)/10], \end{aligned} \tag{3}$$

in which s_i is the semantic similarity appropriate to modality, m_i^t and m_i^v are the indicators of text and image, respectively, ℓ_i is the length of text (zero in the case of images), ℓ_c is the length of the claim and n_c is the number of candidates available. The list of candidates is label-blind, meaning candidates are ranked by s_i, with a limit of 40 text and 20 image items.

C. Evidence-Membership Scorer

The lightweight logistic scorer estimates an uncalibrated inclusion probability

$$q_i = \sigma(w^\top \tilde{x}_i + b), \tag{4}$$

where $\tilde{x}_i$ is obtained by using fitting-fold statistics to standardize it. A class-balanced binary cross-entropy loss is used to address the small number of gold items and fixed L_2 regularization is used for w. The learning rate, the number of epochs and the regularization are determined by the development configuration, and are not changed based on formal-test results.

This scorer is intentionally simple. It is a test of the selection rule, but does not make an architectural-capacity claim. A stronger graph scorer could only be declared as an ablation under the same development budget and could not replace the frozen main scorer after the official test is reviewed.

D. Claim-Grouped Calibration

For each of the predetermined seeds, 20% of the training claim identifiers are deterministically allocated to a calibration fold. The scorer is fit on the remaining claims. A binary temperature parameter is fitted on the calibration probabilities and labels, resulting in

$$p_i = \sigma(\mathrm{logit}(q_i)/T). \tag{5}$$

Brier score, negative log-likelihood, expected calibration error and reliability plots are used separately to assess calibration. Temperature scaling is the default; a more flexible calibrator is not chosen after the fact.

E. Exact Expected-F1 Decoder

The candidates are ranked from highest to lowest calibrated p_i, and if two candidates have the same calibrated p_i, they are ranked according to the order of the input. Let TP_k be the number of gold items within a prefix of length k, and let G be the number of gold items in the whole packet. If the Bernoulli membership variables are independent, then the utility of the prefix is

$$U(k) = \mathbb{E}\left[\frac{2TP_k}{k+G}\middle| p_1, \ldots, p_n\right]. \tag{6}$$

The implementation calculates the distributions of gold counts inside and outside each prefix using Poisson-binomial recurrences and sums their joint mass exactly. It does not substitute a ratio of expectations. The selected size is

$$k^* = \arg\max_{k \in \mathcal{K}} U(k), \qquad S = \{e_{(1)}, \ldots, e_{(k^*)}\}. \tag{7}$$

At least one positive candidate has been released for each MOCHEG development claim in the audited packet, and thus the frozen policy disallows an empty selection. The maximum is 60 items, and the smaller set is used to break ties. Unit tests compare the dynamic program with brute-force enumeration on small probability vectors, and test invalid probabilities, empty inputs, maximum budgets and tie behavior.

F. Matched Controls and Verifier

Fixed selectors return the first k candidates for $k \in \{1, 2, 3, 5, 10, 20, 40, 60\}$. Only permitted calibration data are used to select the best fixed budget. Other controls are a calibrated global threshold, an uncalibrated expected-F1 decoder and a count-classification head. The candidate probabilities and order are the same for all controls.

Each selected evidence set is fed into the shared verifier, which predicts the three MOCHEG labels. This corresponds to one balanced three-class checkpoint per seed, which is reused in all selector outputs, with the same candidate features and decoding inputs. This separates the impact of the chosen set on the verdict diagnostic. A good verifier effect is not able to compensate for a failed selector gate as evidence-set F1 is primary.

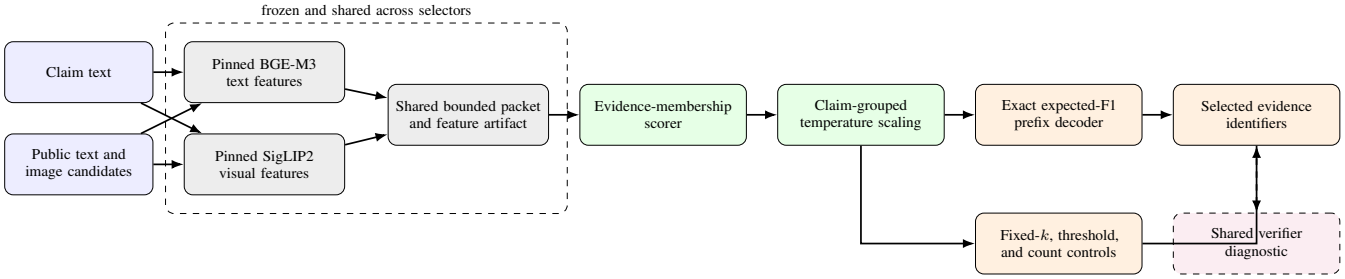


Fig. 1. CEFES development pipeline. Fixed-budget, threshold, count, and CEFES selectors share candidate packets and frozen feature artifacts. The dashed shared-verifier branch is a secondary diagnostic.

V. Experimental Setup

A. Research Questions

The key question is whether CEFES improves claim-level evidence-set F1 compared with the strongest fixed budget selected on development data. Secondary questions investigate whether expected-F1 decoding improves on count prediction, whether calibration improves selection quality, and whether selected evidence preserves verdict macro-F1 when using a shared verifier.

The claim hierarchy is made explicit in Table III. The primary selector claim can only be made based on the first comparison. Calibration, count prediction and verification are diagnostic tools to see why a selector does or does not work, but they all fail to help correct a negative primary evidence effect. This hierarchy was set up to prevent the awarding of whichever secondary endpoint may be favorable.

TABLE III
Prespecified claim-to-test alignment. All comparisons are paired on identical eligible claims.

Question	Comparison	Role
Set utility	CEFES versus strongest permitted fixed k	Primary gate
Count alignment	CEFES versus learned count head	Mechanism control
Calibration	Guarded versus raw probabilities	Safety condition
Verification	Shared-verifier macro-F1 difference	Non-inferiority safeguard

B. Development and Formal Protocols

Seeds 13, 37 and 73 are used for the development pilot. A maximum of 20 declared configurations may be evaluated using train and development data. There is no official test during this phase. If unlocked, a formal evaluation would use seeds 13, 37, 73, 101 and 137 exactly once with the frozen code, data, features, configurations and checkpoints.

The first GPU run was done with a validated 400-claim packet to test the semantic feature path and selector mechanism. It is a feasibility experiment since its candidates are members of the released judgment universe. The main development comparison repeats the selector on the same eligible claim sample after label-blind retrieval from the whole public text and train/development image corpus. Relevance judgments are only added once the rankings are locked. The formal test is not unlocked by a qrel-packet result alone.

C. Baselines and Fairness

The controlled baselines are fixed top-k selection, a calibrated 0.5 probability threshold, uncalibrated expected-F1 decoding and a learned count head. The official baseline for MOCHEG is reproduced as appropriate. The published M-RAV and MEVER values are context-dependent because their candidate construction, model access and tuning conditions differ [4], [5]; they are not put through paired significance testing. All rerun selectors are given the same candidates, features, supervision and configuration budget.

D. Metrics and Uncertainty

The macro precision, recall and F1 are the evidence metrics used at the claim level. The secondary evidence metrics include the exact set match, the Jaccard score, the selected count, the count mean absolute error, Recall@k, MAP and nDCG. Brier score, negative log-likelihood and expected calibration error are measures of calibration. Macro-F1 and class-wise F1 are used to measure verdict quality.

10,000 paired bootstrap resamples are used for main selector effects. The resampling unit is the source, event or fact-check article where reliable grouping metadata are available; otherwise claims are resampled and that limitation is noted. Seed-aligned records are averaged before the main interval, and seed dispersion is reported separately. Holm correction is applied to the prespecified selector family. Verdict preservation is evaluated using a paired non-inferiority interval with margin -0.01.

E. Development Gate

The test only becomes available if all of the following are true: all data and software gates pass; mean CEFES-minus-best-fixed evidence F1 is at least $+0.015$ over the three development seeds; the effect is positive for every seed; the calibrated Brier score is not worse than the uncalibrated one; the verdict macro-F1 is not more than 0.01 below the fixed control; the matched comparisons have the same candidate and verifier artifacts; and the development table is regenerated by a clean rerun. None of the seeds are positive and the mean effect of the full-corpus

selector is negative. Thus, the gate is fail-closed irrespective of secondary diagnostics and the formal-test annotations are not available.

The positive formal claim requires a mean official-test evidence-F1 effect of at least $+0.01$ with a positive 95% paired interval, and the verdict-effect lower bound should be above -0.01. A gain against an arbitrarily weak fixed budget does not satisfy this rule.

F. Ablations and Robustness

Prespecified ablations remove calibration, replace exact expected-F1 with a global threshold or ratio approximation, replace set decoding with count prediction, and remove modality or duplicate-cluster features. Robustness tests involve repeated evidence, varying the width of the packets without retuning, removing each modality and corrupting the top candidates with irrelevant items. Verdict, gold-set size, source and the need for image evidence are covered under subgroup analysis. These analyses explain a primary result but cannot rescue a failed main comparison.

G. Implementation and Resources

The primary execution host is an RTX 4090 with 24 GiB device memory, 64 GiB system memory and an Intel i9 processor with around 1 TiB free storage. The parameters for feature extraction are mixed precision, frozen and cached encoders, a text batch size of 32 and an image batch size of 64. The Python package, configurations, tests, data preparation, feature builder, selector runner and PowerShell orchestration are version controlled. Raw predictions, calibration values, selected identifiers, histories, hashes and environment metadata are stored for independent recomputation.

VI. Results

A. Data and Software Readiness

The evidence available is reported in Table IV. Pinned MOCHEG train and development metadata reproduce the official claim and evidence counts. All 378,864 distinct train/development sentence judgment identifiers are resolved to public corpus text. The deterministic 400-claim packet includes 15,823 candidates, 15,289 available candidates and no unavailable gold candidate. All 1,612 downloaded pilot images are readable.

B. Software-Only Smoke Run

The pipeline was run end to end with a dependency-light hash feature, and the mean difference between CEFES and fixed evidence F1 was $+0.00083$. Its feature manifest explicitly sets `scientific_results_allowed=false`; this means that the selector gate remains false. This value is used to check the serialization, training, calibration, evaluation and export paths only. It is not a model result, does not meet the development criterion and cannot be used in the final abstract as proof of effectiveness.

TABLE IV
CURRENT EVIDENCE STATE. PENDING ENTRIES ARE NOT INTERPRETED AS ZERO OR NEGATIVE PERFORMANCE.

Component	State	Evidence
Train/development identity	Passed	Counts and revisions match
Sentence recovery	Passed	378,864/378,864 identifiers resolved
400-claim packet	Passed	No unavailable gold candidate
Exact decoder	Passed	Dynamic program agrees with brute force
Project test suite	Passed	98 tests after preservation repair
Pinned semantic GPU pilot	Checked	Three seeds; independently recomputed
Matched verifier	Checked	One shared checkpoint per seed
Full-corpus retrieval	Checked	91,822 text documents; 90,270 image paths
Clean reproduction	Passed	Features, rankings, and method tables match
Official test	Sealed	No active-study test annotation accessed

C. Scientific Performance

The BGE-M3/SigLIP2 feature artifact has 15,289 available claim–candidate pairs, 4,156 of which are gold rows. Feature construction took 626.16 seconds and peaked at 1.71 GB of allocated GPU memory. The first temperature-scaling trial increased mean evidence F1 over the selected fixed budget but failed its calibration condition due to an increase in Brier score for seed 37 of 0.00020. This unsuccessful trial remains.

The second, announced trial is a claim-disjoint training guard that picks either a fitted temperature or an identity scaling. It independently recalculates its results and reports them in Table V. Mean CEFES F1 is 0.71978 versus 0.65145 for fixed selection, a difference of 0.06833. Guarded Brier score is no worse in all seeds and the effect is positive in all seeds. However, the calibrated 0.5-threshold control has F1 values of 0.73351, 0.75153 and 0.73979, which are better than CEFES for all seeds. So the criteria for a fixed-budget selector hold and the evidence does not warrant that the strongest rule available is one for exact expected-F1 decoding.

TABLE V
CHECKED 400-CLAIM DEVELOPMENT SELECTOR RESULTS. INTERVALS ARE 10,000-SAMPLE PAIRED CLAIM BOOTSTRAPS.

Seed	CEFES	Fixed	Difference	95% interval
13	0.71248	0.65313	+0.05935	[0.04146, 0.07779]
37	0.72091	0.65603	+0.06489	[0.04551, 0.08429]
73	0.72595	0.64519	+0.08076	[0.05485, 0.10677]

The shared verifier uses one balanced three-class checkpoint per seed for every selector. The checked CEFES-minus-fixed macro-F1 changes are -0.01594, $+0.00900$, and $+0.00656$ for seeds 13, 37, and 73. These have a mean of -0.00013, which is above the declared non-inferiority margin of -0.01,

but all paired intervals contain 0 and seed 13 is individually below the non-inferiority margin. This supports neither a verdict improvement nor equivalence; it merely passes the prespecified mean point-margin check.

Independent audits revealed no differences in the hashes of the artifacts, probabilities, selections, evidence/verdict measures or bootstrap samples. The selector and verifier archives have SHA-256 values of `5d62d125...22b8e` and `e8491d7e...abbdc`, respectively. These feasibility results motivated but did not unlock the full-corpus comparison.

D. Full-Corpus Development Comparison

The label-blind builder ranks 91,822 released text documents in the first stage, reranks sentences from the top 50 documents and ranks all 90,270 train/development image paths independently. It stores 40 text and 20 image candidates per eligible claim before joining qrels. The resulting artifact contains 21,972 candidate rows for 367 claims and has SHA-256 `68889ae3...590731f`. It took 1,644.45 seconds to build and used up to 1.75 GB of GPU memory. The complete-gold retrieval metrics and artifact identities were independently recalculated, with no differences found. The main comparison is thus under significant retrieval error: text Recall@50 = 0.36160 and image Recall@50 = 0.21384.

The experimental reversal is summarized in Table VI. The packet experiment and full-corpus comparison use the same family of selectors and fixed semantic encoders but answer different questions. The former asks whether the decoder can choose from an assessed universe of candidates; the latter asks whether the entire retrieval–selection chain can support the claimed advantage. This change of sign is thus not an inconsistent repetition of measurement. It is evidence that the conclusion depends on how candidates enter the selector.

TABLE VI
COMPARISON OF THE TWO CHECKED DEVELOPMENT REGIMES. CANDIDATE ROWS ARE THE AVAILABLE RECORDS SUPPLIED TO THE SELECTOR.

Property	Judgment packet	Full corpus
Eligible claims	400	367
Candidate rows	15,289	21,972
Candidate origin	Released judgments	Label-blind retrieval
Search universe	Assessed items	91,822 documents and 90,270 images
Gold availability	All retained claim gold available	Limited by retrieval
Text/image Recall@50	Not an end-to-end retrieval measure	0.36160 / 0.21384
Mean CEFES–fixed F1	$+0.06833$	-0.04185
Stronger matched control	Threshold	Fixed and count
Scientific role	Feasibility	Primary development gate

Table VII lists the result of the audited selector. CEFES is beaten by the fixed budget that the validation selects in all seeds. The mean difference is -0.04185 and all paired intervals are negative. The calibration safety guard chooses identity scaling in each seed, and thus calibrated and raw Brier scores are equal and the non-worsening condition is satisfied. However, the prespecified mean-gain and all-seed-positive conditions still fail.

TABLE VII
CHECKED FULL-CORPUS DEVELOPMENT SELECTOR RESULTS. INTERVALS ARE 10,000-SAMPLE PAIRED CLAIM BOOTSTRAPS.

Seed	CEFES	Fixed	Difference	95% interval
13	0.08210	0.12189	-0.03978	$[-0.05218, -0.02744]$
37	0.08170	0.12869	-0.04699	$[-0.06659, -0.02807]$
73	0.08178	0.12055	-0.03877	$[-0.05102, -0.02683]$

The matched count-classification control gives mean evidence F1 0.10040 whereas CEFES gives 0.08186. All CEFES-minus-count differences are negative, with a mean difference of -0.01854, and all three paired intervals exclude zero below zero. Count prediction itself is inaccurate (count mean absolute error 16.15–17.87), indicating that it can still outperform expected-F1 decoding on this ranking without estimating the true set size well.

The shared-verifier diagnostic gives a mean CEFES-minus-fixed macro-F1 difference of $+0.05768$. The point differences for the three seeds are $+0.04855$, $+0.05453$, and $+0.06995$, with only seed 73's paired interval excluding 0. This diagnostic passes the declared mean point non-inferiority check, but cannot rescue the failed primary evidence conditions. The independent audits of the selector, the count control and the verifier report that there were no discrepancies and no formal-test access.

A clean end-to-end rerun independently rebuilt the feature store in 1,621.25 seconds. The compressed feature SHA-256 is exactly the same as in the first run, as are the sidecar, candidate identities, complete-gold maps, rankings, all selector and control metrics, and bootstrap samples within the specified tolerance. The fail-closed gate meets the data, software, verifier, calibration, matched-artifact and clean-reproduction conditions, but does not meet the minimum mean evidence gain or positive-effect-in-every-seed conditions. The development archive of 74 members is frozen with a SHA-256 of `b7961df9...5c21d00b` and records `formal_test_unlocked=false`.

The official test has not been accessed for the current study. As a consequence, no formal superiority, robustness or generalization conclusion is reported in this draft.

VII. DISCUSSION

A. Why the Candidate Regime Changes the Conclusion

Using the checked packet experiment and the full-corpus comparison gives opposite results. Adaptive calibrated selection is substantially superior to the best development-selected fixed budget on the released judgment-candidate packet, although a global calibrated threshold is stronger in every seed. However, under label-blind full-corpus retrieval, CEFES performs worse than the selected fixed budget in every seed. The candidate universe is therefore easier for the packet result, which does not establish end-to-end retrieval effectiveness.

One step in this change is accounted for by the reachability bound in Eq. 2. A selector cannot recover an item that retrieval

did not put in its packet. Low text and image Recall@50 lower the potential evidence score before calibration or decoding. The remaining candidates are also distributed differently: assessed items in a judgment packet are related to the claim, while high-similarity distractors in a full-corpus packet are drawn from tens of thousands of documents and images. This means that the evidence-membership scorer faces both missing positives and a harder negative distribution.

A second qualification is added by the threshold result. Calibrated thresholding outperforms exact expected-F1 decoding in every seed on the judgment packet, even though both methods use the same probabilities. While this packet therefore supports adaptive set sizing in a general sense, it does not identify the expected-F1 decoder as the strongest adaptive rule. On the full corpus, the count head exceeds CEFES despite poor count mean absolute error. This indicates that count accuracy and set F1 are still not the same, and that directly decoding expected utilities is not necessarily robust to a poor ranking.

B. What the Negative Result Establishes

The results of the calibration and control show the reason for the conjunctive nature of the gate. Unrestricted temperature scaling slightly worsens the Brier score for one packet seed despite the positive evidence-F1 effect. The declared training-only guard repairs that failure without consulting validation labels. The full-corpus run retains identity scaling, yet expected-F1 decoding is still outperformed by the fixed-budget and count-classification controls. The shared verifier has a favorable mean point estimate, but this secondary verdict effect does not replace evidence quality.

The formal benchmark can be consumed only once for a confirmatory interpretation, thus making the development gate a demanding one. It also avoids the primary evaluation mistake of the previous count-prediction study: beating one convenient fixed budget while losing to another sensible budget. Here it is effective as intended: even with a positive feasibility packet and a favorable verifier diagnostic, the cleanly reproduced negative full-corpus effect stops the study before the formal test.

The result does not imply that expected-F1 decoding is inherently inferior. Full-corpus Recall@50 for text and images are only 0.3616 and 0.2138, respectively, and item-membership estimates might not satisfy the conditional-independence assumption of the decoder. It does establish that the declared method, scorer and retrieval configuration do not justify consuming MOCHEG's formal test. Future directions should be treated as new studies with a new development protocol, not an a posteriori rescue of this one.

This boundary is important when assessing a negative development decision. The experiment does not prove that all adaptive selectors, scorers or retrievers will fail; it falsifies the prespecified superiority claim for the evaluated configuration. On the other hand, the favorable packet and verifier diagnostics are not applicable outside the conditions in which they were measured. The checked conclusion is intentionally more limited: with the pinned BGE-M3/SigLIP2 retrieval artifacts, logistic evidence scorer, calibration guard and declared decoder, there is no primary evidence advantage, and the effect is consistently negative across the three development seeds.

C. Implications for Evaluation Practice

First, candidate construction needs to be reported as part of the evaluated system. A selector-only experiment on judgment candidates is helpful for mechanism development, but it should be identified as conditional on candidate availability. End-to-end claims require label-blind retrieval and complete-gold scoring to prevent missed evidence from being silently removed from the denominator. This distinction can be made apparent by reporting retrieval Recall@k and the packet-conditioned ceiling in addition to selector F1.

Second, selector comparisons require a common packet and a strong fixed comparator. Permitting methods to receive different candidates confounds retrieval with set decoding. After observing results, it is possible to choose a weak fixed budget, which results in an artificial adaptive gain. In this study candidate identities and features are shared within each regime, the fixed budget is chosen using permitted calibration data, and all declared seeds are preserved. These controls help to interpret the negative effect, but do not make it better.

Third, evidence selection and verdict prediction should remain separate endpoints. A verifier may benefit from additional context that is not part of the released gold evidence, or it may use correlations that do not support faithful attribution. Thus the positive mean verifier diagnostic is relevant to system behavior but cannot establish evidence quality. A credible multimodal fact-checking evaluation should report both outcomes and specify in advance which one governs the primary claim.

Finally, staged evaluation is most informative when all the various stages keep their failures. This is a unified record of the unsuccessful temperature-scaling attempt, the more potent threshold control, the negative full-corpus seeds and the good verifier diagnostic. Saving all of them helps to keep a positive feasibility result from being separated from what led to the reversed result, and provides future researchers with tangible hypotheses to work into a different protocol in terms of retrieval coverage, calibration, and evidence dependence.

VIII. Limitations and Ethical Considerations

This manuscript reports a completed development decision, not a formal-test result. The 400-claim feasibility packet begins with released judgment candidates and clearly overestimates performance relative to the label-blind full-corpus comparison. There are 367 eligible claims in the development sample after the fixed exclusion policy, rather than all released train/development claims. Conditionally independent evidence-membership variables are assumed even though documents and duplicated images may be dependent. A poorly ranked group of candidates cannot be corrected by temperature scaling, and the simple logistic scorer is not a capacity measure. Since the primary development gate failed, robustness tests, the five-seed formal matrix and external validation were not performed. MOCHEG includes inherited professional fact-check labels and English-language web evidence, so results may not be comparable to other languages, domains or live information environments. Public mirrors reproduce the official counts but remain secondary transports; claim-level uncertainty is used because reliable event grouping is unavailable. All these

limitations preclude any assertion of deployment readiness or a general claim to fact-checking reliability.

The system is designed to aid in evidence review, not to automatically decide on the truth. Verdicts or evidence may be wrong or incomplete and result in misrepresentation of people, events and sources. The project leverages public research information, records provenance and immutable identifiers but does not pass on credentials or private user information. Fluently written explanations do not count as evidence of proper attribution and are therefore not included in the main claim. Failure results, selected identifiers, hashes, and probabilities must be retained as formal outputs to allow the unfavorable results to be auditable.

IX. Conclusion

We test an adaptive selector, CEFES, which selects a multimodal evidence prefix using exact expected set-F1 computed from calibrated item probabilities. A positive 400-claim judgment-packet result reverses under label-blind full-corpus retrieval: CEFES underperforms the validation-selected fixed budget and a matched count-classification control in every seed. A clean rerun regenerates the feature store, rankings, method tables and bootstrap results, and independent audits find no discrepancies. The prespecified primary evidence conditions fail, though a shared-verifier diagnostic is favorable. The complete negative development record is frozen, the official test remains sealed and no claim of superiority can be made.

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